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Shock metamorphism and petrography of the Shergotty achondrite*

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Abstract—The Shergotty subsamples ,1 and ,12 consist of augite and pigeonite (67.5%), maskelynite (24%), ilmenite and titanomagnetite (2%), pyrrhotite (0.4%), whitlockite (1.8%), apatite (0.1%), quartz (0.5%), baddeleyite (trace), fayalite (0.4%), mesostasis (3%), and shock-induced local, polymineralic melt products (0.6%). The overall modal composition is similar to other Shergotty samples except for the rather high whitlockite content. The shock effects observed in the mineral constituents include mosaicism, deformation bands, planar fractures, and mechanical twin lamellae in clinopyroxene; isotropization of plagioclase with very rare remnants of birefringence; planar deformation structures, mosaicism, and strongly reduced birefringence in quartz; mechanical twinning of ilmenite; localized *in situ* melting of neighbouring minerals at the contact of low and high density phases. Based on the refractive index of maskelynite (average: 1.5467 with average An-content of 49%) and the degree of isotropization of the plagioclase an equilibrium shock pressure of 29 ± 1 GPa is derived. The inferred post-shock temperature is 200 ± 20°C. No heating event could have exceeded 400°C (DUKE, 1968). Local stress and temperature concentrations reach 60–80 GPa and 1600–2000°C. The observed shock effects can be explained by a single shock event. A second, weaker shock event as found by others appears to be highly improbable. Equilibrium shock pressures and post-shock temperatures for the other known shergottites are 31 ± 2 GPa and 220 ± 50°C (Zagami), 43 ± 2 GPa and 400–800°C (ALHA 77005). The pressure estimate for EETA 79001 by LAMBERT (1985) is confirmed: 34 ± 1 GPa; the post-shock temperature is 250 ± 50°C. The abundance and textural setting of localized melt products in these meteorites confirm increasing shock pressures in the sequence Shergotty, Zagami, EETA 79001, ALHA 77005. Undoubtedly, the melts could have been formed by the same single shock which produced the equilibrium shock effects (*e.g.* maskelynitization) in these meteorites. The shock-induced particle velocities inferred from Hugoniot data of basalts are in the 1.5–2.0 km/s range for the parental rocks of the shergottites. Ejection velocities are therefore in the order of 3–4 km/s. Special ejection mechanisms are required in order to exceed the escape velocity of a planet like Mars without producing higher degrees of shock (*e.g.* melting) than those observed in the shergottites.

INTRODUCTION

THE STUDY OF the Shergotty achondrite as a member of the Shergottites-Nakhlites-Chassigny Group (SNC) is related to two major problems in the scope of its possible Martian origin (*e.g.*, WOOD and ASHWAL, 1981): (1) Its chemical and mineralogical composition is pertinent to the composition and age of Martian basalts and thereby provides insight into the composition and evolution of the Martian mantle and the planet as a whole (WOOD and ASHWAL, 1981; SHIH *et al.*, 1982; DREIBUS *et al.*, 1982; BOGARD *et al.*, 1984; DREIBUS and WÄNKE, 1984; SMITH *et al.*, 1984); (2) Its shock history is pertinent to the ejection dynamics for rock fragments escaping the gravity field of Mars (NYQUIST, 1983, 1984; MELOSH, 1984; WETHERILL, 1984).

Shergotty is a non-brecciated, shocked basalt of diabase type with cumulate texture (Fig. 1). The degree of shock can be classified as stage II according to STÖFFLER (1971a, 1984) or class 2 according to KIEFFER *et al.* (1976) because of the presence of diaplectic plagioclase glass (maskelynite). Both petrography and shock metamorphism have been studied repeatedly since the classic investigations of TSCHERMAK (1872) and MICHEL (1912). More recent petrologic data were published by DUKE (1968), SMITH and HERVIG (1979), STOLPER and MCSWEEN (1979), STOLPER *et al.* (1979), and STOLPER (1979). Although BINNS (1967), DUKE (1968), BUNCH *et al.* (1968), LAMBERT (1983), and LAMBERT and GRIEVE (1984) described the shock effects in Shergotty and discussed their significance, a detailed study of the shock metamorphism of Shergotty and other members of the SNC group is still lacking.

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FIG. 1. Thin section ,34 of the Shergotty meteorite; longest dimension 1.6 cm.

This paper has its main emphasis on a quantitative evaluation of the shock history of Shergotty but also includes—as requested by the leader of the Shergotty consortium—the main petrographic and mineralogical characteristics of the new Shergotty subsamples which were made available to this consortium study. The latter data were mainly provided by the last three authors.

SAMPLES AND METHODS OF INVESTIGATION

We have analyzed thin sections of both subsamples ,1 (chip C) and ,12 (chip B) which were removed from the main sample supplied by the Geological Survey of India (LAUL, 1986). Thin sections ,28; ,29; ,32; and ,33 were available to Simon and Papike, thin sections ,29 and ,33 to Sen Gupta (see SEN GUPTA, 1985), and thin sections ,30 and ,34 to the remaining authors. An ultra-thin section of chip B was studied by Beauchamp and by the Münster group (see LAUL, 1986).

Refractive index measurements were made on a 40–100 μm grain fraction of maskelynite obtained by E. Jagoutz, Mainz. A spindle-stage technique (MEDENBACH, 1985) was applied which yields a precision of ±0.0003. Microprobe analyses were performed at the laboratories of Rapid City, Münster, and Mainz (Max-Planck-Institut für Chemie). The instruments and measuring conditions were: ARL SEMQ-51 microprobe, wavelength dispersive analysis at 15 kV and 20 nA on brass (Münster), ARL SEMQ-51 microprobe, energy dispersive analysis with a KEVEX system at 15 kV and 6 nA on copper (Mainz), and MAC-5 microprobe, wavelength dispersive analysis at 15 kV and 15 nA on brass (Rapid City).

PETROGRAPHY AND MINERAL CHEMISTRY

The mineral constituents which were identified in the thin sections of chips B and C of the Shergotty sample from the Geological Survey of India and their abundances are summarized in Table 1. The overall textural relationships among these minerals can be seen from the photograph of a whole thin section (Fig. 1). Results of microprobe analyses of various minerals are plotted in Fig. 2. Selected analyses of pyroxene, maskelynite, whitlockite, and a silica-phase are given in Tables 2, 3, and 4.

Figure 1 reveals the typical igneous texture of Shergotty which is dominated by prismatic augite and pigeonite (up to 4 × 0.8 mm, rarely up to 6 mm in size) forming a cumulate texture in which subhedral to anhedral lathy or stubby maskelynite (up to 2 mm in length) fills the interstitial space as an intercumulus phase together with the accessory minerals and mesostasis. In some places, pyroxene and plagioclase are interfingered or plagioclase encloses small grains of pyroxene. The thin sections studied here do not show the well-known foliation of Shergotty (DUKE, 1968), probably because of the unsuitable orientation of the cuts. Point counting in two perpendicular directions on each of the sections ,28 and ,32 did not reveal any preferred orientation. The modal compositions of Table 1 indicate slightly less pyroxene compared to previous analyses obtained on other samples of Shergotty. More distinct differences exist for the abundance of accessory minerals. In particular, the present samples are richer in whitlockite than other samples (Table 1).

Table 1: Modal composition of Shergotty samples obtained by point-counting

	1	2	3	4	5	6	7	8	9	Chip C average 4, 5, 9	Chip B average 3, 6
Clinopyroxene	73.4	75	65.5	67.8	70.1	66.2	70.5	69.1	67.7	68.5	65.9
Maskelynite	22.5	20	27.2	24.1	22.8	25.4	23.9	22.7	23.8	23.6	26.3
Mesostasis	-		2.8	3.6	3.1	3.2	2.8	5.2	2.4	3.0	3.0
Magnetite	4.5	2	2.1	2.2	2.0	2.1	2.0	2.5	} 1.6	} 2.0	2.1
Ilmenite	-	} 1	0.2	tr	0.1	tr	0.5	0.3			tr
Pyrrhotite	-		0.3	0.5	0.1	0.5	0.3	0.2	0.7	0.4	0.4
Whitlockite	-	1	1.9	1.7	1.6	2.4	tr	tr	1.4	1.6	2.2
Fayalite	-	-	tr	-	0.2	0.2	tr	tr	0.4	0.2	0.1
Apatite	-	-	-	tr	-	-	-	-	0.1		
Quartz	-	1	-	-	-	-	-	-	0.5		
Baddeleyite	-	-	-	-	-	-	-	-	tr		
Shock melt	-	-	-	-	-	-	-	-	0.6		
SiO_2-K-rich inclusions in pyroxene and whitlockite	-	-	-	-	-	-	-	-	0.7		
n	?	?	2000	2000	2000	2000	1274	1300	2239		

1 = Tschermak, 1872; 2 = Duke, 1968; 3 = thin section ,32; 4 = thin section ,28; 5 = thin section ,29; 6 = thin section ,33; 3-6 = Simon et al., 1985; 7 and 8 = thin sections from Shergotty USNM 321, Stolper and McSween, 1979; tr = trace; n = number of counted points; baddeleyite observed by Smith and Hervig (1979) also; 9 = thin section ,30 (Stöffler)

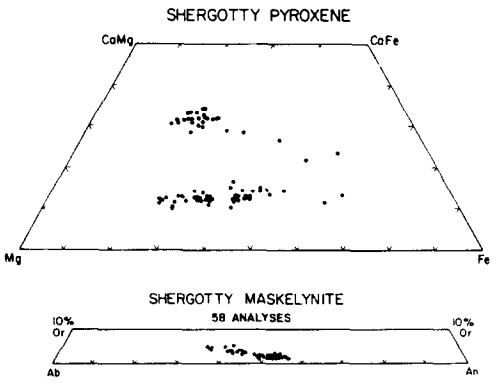


FIG. 2. Composition of pyroxenes and maskelynite in Shergotty thin section ,32.

Most of the petrographic and mineralogical characteristics reported previously (DUKE, 1968; SMITH and HERVIG, 1979; STOLPER and MCSWEEN, 1979) such as twinning and zoning of pyroxene, zoning of maskelynite, intergrowth of Ti-magnetite and ilmenite, reaction rims of fayalite on magnetite, and two-phase composition of mestostasis with a silica- and an alkali feldspar-like phase are observed in our thin sections as well. Also the results of microprobe analyses (Tables 2, 3) confirm previous findings (STOLPER and MCSWEEN, 1979; SMITH and HERVIG, 1979). Two types of pyroxene, pigeonite and augite, reveal the compositional trends shown in Fig. 2; their compositional ranges are approximately Wo_{10-17} $En_{64-23}Fs_{24-63}$ and $Wo_{22-34}En_{52-20}Fs_{18-56}$, respectively. The maskelynite composition ranges from $Or_{4.5}Ab_{59.5}An_{36}$ to $Or_{1.2}Ab_{43}An_{56}$. This compares with the An_{41}-An_{57}-range observed by SMITH and HERVIG (1979) and STOLPER and MCSWEEN (1979). In a weakly birefringent, slightly brownish mineral also described by MICHEL (1912) and DUKE (1968) we observed planar deformation structures not mentioned previously. This mineral forms relatively large interstitial grains, up to 200 μm in size, and consists of nearly pure silica (96–97 wt. % SiO_2; Table 4). The relatively high Al_2O_3- and alkali-contents may favour shocked tridymite over shocked quartz, although cristobalite (secondary?) has been identified in these grains by DUKE (1968). However, X-ray diffraction analysis of single grains with a Gandolfi camera revealed the two strongest lines of α-quartz only.

In summary, we conclude that the newly available Shergotty sample is quite similar to previously analyzed samples. Consequently, the results of the present Shergotty consortium study should be applicable to the Shergotty meteorite in general.

SHOCK METAMORPHISM

The shock effects observed in the Shergotty samples can be classified into two basic types: (1) mechanical deformations and transformations which take place essentially in the solid state; and (2) localized melting at grain boundaries of high density and low density minerals yielding melts with a very fine grained crystalline texture. The former type includes planar deformation structures, mechanical twinning, and formation of diaplectic plagioclase glass (maskelynite). The localized, *in situ* formed melt products which have not been described in detail before result from partial melting of neighbouring mineral grains such as pyroxene, maskelynite, pyrrhotite, titanomagnetite. The types of shock effects found in the consortium samples are summarized in Table 5 and described below.

(1) Solid state shock effects

Plagioclase, ranging from An_{36} to An_{56} in the thin sections investigated (LAUL, 1986, and Fig. 1), is completely transformed to diaplectic glass in most grains although remnants of strongly reduced birefringence have been observed in very few grains. This observation holds also for other sections of the Shergotty meteorite (TSCHERMAK, 1872; MICHEL, 1912; BUNCH *et al.*, 1967; DUKE, 1968). Weakly birefringent relics are only mentioned by BUNCH *et al.* (1967). All morphological features of the pre-shock crystalline state such as grain boundaries, cleavage, zoning, twin boundaries, and delicate inclusions are preserved in the isotropic state. Only rarely, offsets of fractured apatite needles were observed. Any sign of vesiculation or flowage of the maskelynite is definitely lacking (Fig. 3).

Pyroxene displays an extremely pervasive mosaicism as indicated by a pronounced irregular optical extinction on a very small scale (Fig. 4). Occasionally, deformation bands are developed (Fig. 4). The observed type of mosaicism which can be best seen in the ultrathin section (chip B) is characteristic of the degree of shock metamorphism (compare HÖRZ *et al.*, 1986) which accounts for the maskelynitization of coexisting plagioclase (see STÖFFLER, 1972, for definition of shock-induced mosaicism). All pyroxene grains are thoroughly fractured (Figs. 3 and 5). The fractures are predominantly irregular and non-planar, some cut through whole crystals, others are limited to small areas inside the crystals or extend from grain boundaries or from points in a radial pattern. Planar fractures which obviously represent the characteristic cleavage of pyroxene are also common. The crystal regions between fractures are deformed by parallel or subparallel sets of planar elements of different crystallographic orientation. These planar elements (see STÖFFLER, 1972, for definition) appear as very thin optical discontinuities with an irregular spacing of about 2–10 μm. Their frequency and density increase in areas adjacent to *in situ* melt products (see below). Such planar elements are well-known from pyroxene of shocked terrestrial, lunar and meteoritic rocks (STÖFFLER, 1972 and references therein). A fourth type of unequivocal shock deformation is polysynthetic mechanical twins subparallel to (001). Under the microscope

Table 2: Analyses of Shergotty Pyroxene

1	2	3	4	5	6	
SiO_2	53.50	50.84	48.16	52.21	52.39	50.69
Al_2O_3	0.63	0.98	1.10	0.88	1.07	1.02
FeO	15.30	25.66	35.70	13.87	12.34	17.94
Fe_2O_3	0.39	0.43	0	0.91	0	1.51
MgO	22.72	14.90	7.54	16.87	15.94	13.64
MnO	0.57	0.90	0.99	0.55	0.59	0.59
TiO_2	0.10	0.24	0.54	0.20	0.21	0.39
Cr_2O_3	0.52	0.19	0.07	0.61	0.47	0
CaO	6.18	5.83	5.83	13.86	16.16	13.62
Na_2O	0.06	0.09	0.02	0.22	0.14	0.27
Total	99.97	100.06	99.95	100.18	99.31	99.67
			Cations per 6 oxygens			
Si	1.968	1.965	1.961	1.954	1.970	1.948
Al	0.027	0.045	0.053	0.039	0.047	0.046
Fe^{2+}	0.471	0.829	1.216	0.434	0.388	0.576
Fe^{3+}	0.011	0.012	0.000	0.025	0.000	0.044
Mg	1.245	0.858	0.458	0.941	0.893	0.781
Mn	0.018	0.029	0.034	0.017	0.019	0.019
Ti	0.003	0.007	0.017	0.006	0.006	0.011
Cr	0.015	0.006	0.002	0.018	0.014	0.000
Ca	0.244	0.241	0.254	0.556	0.651	0.561
Na	0.004	0.007	0.002	0.016	0.010	0.020
Total	4.006	3.999	3.997	4.006	3.998	4.006
Wo	12.4	12.5	13.2	28.8	33.7	29.2
En	63.6	44.5	23.7	48.7	46.2	40.7
Fs	24.0	43.0	63.1	22.5	20.1	30.1

thin sections ,28 and ,32

Table 3. Analyses of Shergotty Maskelynite

	1	2	3	4	5	6	7	8	9	10
SiO_2	55.45	56.35	58.33	59.06	56.97	57.69	55.52	55.23	55.58	53.95
Al_2O_3	27.90	26.52	25.76	25.37	26.93	27.33	28.14	28.15	27.27	27.85
FeO	0.73	0.71	0.99	0.66	0.83	0.46	0.56	0.53	0.72	0.52
MgO	n.d.	n.d.	n.d.	n.d.	0.08	0.10	0.10	0.16	0.07	0.08
CaO	10.88	10.83	8.73	6.80	10.75	9.88	10.82	10.72	10.33	10.78
Na_2O	4.77	5.44	5.82	6.25	5.41	5.51	5.02	4.97	5.49	4.81
K_2O	0.31	0.25	0.41	0.72	0.26	0.35	0.21	0.16	0.28	0.32
Total	100.04	100.10	100.04	98.86	101.36	101.32	100.37	99.92	99.74	98.31
	Cations per 8 oxygens									
Si	2.502	2.544	2.617	2.664	2.540	2.560	2.495	2.491	2.516	2.479
Al	1.484	1.411	1.363	1.349	1.413	1.427	1.487	1.493	1.453	1.506
Fe	0.028	0.027	0.037	0.025	0.030	0.017	0.021	0.020	0.027	0.020
Mg	n.d.	n.d.	n.d.	n.d.	0.067	0.067	0.007	0.011	0.047	0.005
Ca	0.526	0.524	0.420	0.329	0.513	0.470	0.521	0.518	0.501	0.531
Na	0.417	0.476	0.506	0.547	0.467	0.473	0.436	0.433	0.481	0.428
K	0.018	0.014	0.023	0.041	0.013	0.020	0.012	0.009	0.016	0.019
Total	4.975	4.996	4.966	4.955	5.043	5.034	4.979	4.975	5.041	4.988
Or	1.9	1.4	2.4	4.5	1.5	2.07	1.24	0.98	1.59	1.91
Ab	43.4	46.9	53.3	59.6	46.94	49.18	45.06	45.14	48.28	43.82
An	54.7	51.7	44.3	35.9	51.56	48.75	53.71	53.88	50.18	54.27

1 - 4: thin sections ,28 and ,32; 5 - 10: thin section ,30 ; n.d. = not determined

the twins appear as sets of narrow, parallel or slightly lense-shaped bands of about 1–5, rarely 10 μm width (Figs. 3 and 4). They may be mistaken as exsolution lamellae, however, shock-induced lamellae of this type have been produced experimentally in diopside (HORNEMANN and MÜLLER, 1971) and bronzite (GIBBONS, 1974) over a wide range of shock pressures (5–53 GPa). They are also ubiquitous in naturally shocked, terrestrial and extraterrestrial pyroxene (STÖFFLER, 1972). In addition, exsolution lamellae may be present although microprobe traverses did not reveal compositional differences within those sets of lamellae which were checked by the Münster group.

Silica, in the form of strongly shocked primary quartz, has been found adjacent to maskelynite as rare, relatively large grains (up to 200 μm in size) which display extremely low birefingence, strong mosaicism, and multiple sets of closely spaced planar elements (Fig. 6). X-ray analysis of single grains revealed diffraction lines of α-quartz. The planar elements which were best visible in the ultrathin section belong to the type of "non-decorated planar elements" (ENGELHARDT and BERTSCH, 1969; STÖFFLER, 1972). The crystallographic orientation of the planar elements could not be measured. However, their density and multiplicity of differently oriented sets are characteristic of quartz which was shocked to about 30 GPa (REHFELDT, 1986, unpublished data) and which corresponds probably to type E of ROBERTSON *et al.* (1968). This type of quartz coexists with diaplectic feldspar glass in naturally shocked terrestrial quartzofeldspathic rocks of shock stage II according to our classification (STÖFFLER, 1971a; see also CHAO, 1968; DENCE, 1968; and STÖFFLER, 1967, 1972, 1984). A fine grained silica phase also occurs in the mesostasis of Shergotty together with an alkali feldspar phase (DUKE, 1968; SMITH and HERVIG, 1979; STOLPER and MCSWEEN, 1979). We were not able to detect unequivocal shock effects in these fine grained phases.

In one of the localized melt pockets of Shergotty described below, a large rounded grain composed of silica with an optical reflectivity similar to pyroxene was found. It is highly birefringent and displays a recrystallized texture. The nature of this mineral cannot be clarified without further analysis; how-

Table 4: Average whitlockite composition and composition of silica phase in Shergotty; 1 = whitlockite of thin section ,29; 2 - 6 = birefringent silica phase with planar deformation structures in ultrathin section of chip B; 7 and 8 = single, highly reflective, polycrystalline grain in melt within pyroxene of thin section ,30; n = number of analyses; σ = standard deviation

Oxide	1	2	σ	3	σ	4	σ	5	σ	6	σ	7	8
SiO_2	0.13	97.2	1.52	96.6	0.9	96.7	1.2	96.4	0.8	96.1	0.6	97.55	97.53
Al_2O_3	0.05	2.05	0.07	1.94	0.15	1.91	0.07	1.98	0.04	2.17	0.09	1.26	1.09
FeO	4.23	0		0		0		0.03	0.06	0		0.20	0.14
MgO	1.29	0.17	0.03	0.14	0.10	0.12	0.04	0.18	0.04	0.15	0.05	0.10	0.04
MnO	0.14	0		0		0		0		0		0	0
CaO	42.01	0.23	0.04	0.14	0.10	0.22	0.03	0.20	0.03	0.24	0.03	0.12	0.09
Na_2O	0	0.92	0.08	0.95	0.14	0.80	0.13	0.80	0.17	0.93	0.16	0.76	0.48
K_2O	0	0.05	0.01	0.05	0.04	0.01	0.03	0.03	0.04	0.03	0.04	0.03	0
TiO_2	0	0.23	0.05	0.18	0.06	0.14	0.02	0.19	0.02	0.20	0.03	0.12	0.21
Cr_2O_3	0	0		0		0		0		0		0	0.04
P_2O_5	44.30	0.11	0.08	0.05	0.06	0.04	0.08	0.03	0.05	0.03	0.06	0	0
Y_2O_3	0.13												
La_2O_3	0.14												
total	92.42	100.96		100.05		99.94		99.84		99.85		100.14	99.62
n		4		4		4		4		4		2	1

Table 5: Shock effects observed in various mineral constituents of the Shergotty meteorite; ( ) = rarely observed

Shock effect	Plagio-clase	Clino-pyroxene	Quartz	Ti-magnetite	Ilme-nite	Pyrrho-tite
Strongly reduced birefringence	(+)		+			
Diaplectic glass	+					
Mosaicism		+	+			
Fracturing	(+)	+	+	+	+	+
Planar elements		+	+			
Mechanical twins		+			+	(+)
Localized melting	+	+	(+)	+	+	+

silica phase = quartz

ever, it is possible that it represents a high pressure polymorph of silica, perhaps stishovite.

Titanomagnetite and *ilmenite*, commonly intergrown, are fractured irregularly. Shock induced mechanical twins are abundant in some ilmenite grains (Fig. 7). Twinning of ilmenite parallel to $\{10\bar{1}1\}$ and (0001) is known from terrestrial and lunar shocked ilmenite (MINKIN and CHAO, 1971).

(2) Liquid state shock effects

In a few very localized regions of thin sections ,30 and ,34 dark brown melt products have been observed which are irregular to rounded in shape and about 0.2–1 mm in size (Fig. 8). They are confined to grain boundaries of pyroxene and opaque minerals (pyrrhotite and titanomagnetite) mainly, but also maskelynite and accessory minerals have been found adjacent to the melted regions. In some cases the melts appear to penetrate into fractures of the host minerals such as pyroxene (Fig. 8). Usually, the minerals close to the melted regions show smoothly curved margins at the contact to the melts indicating that they were incompletely molten and that the melts were formed *in situ* (Figs. 9 and 10). The frequency and density of planar deformation structures in pyroxene increases towards the melt pockets indicating a local increase of shock intensity.

The melt products are heterogeneous in texture and composition. They consist of a very fine-grained, polycrystalline silicate matrix in which spherical or subrounded micron-sized bodies of iron sulfide are embedded. Most obviously, the sulfide globules formed by exsolution of sulfide melt from the silicate melt due to immiscibility (Figs. 8 and 10). Chemically, the melt products represent a mixture of clinopyroxene, plagioclase, phosphates, Fe-Ti-oxides (titanomagnetite and ilmenite), Fe-sulfide (pyrrhotite) and probably also silica and K-rich phases (Table 6).

The observed localized melt pockets of the Shergotty meteorite are morphologically different from pseudotachylite veins frequently observed in shocked lunar and terrestrial rocks (STÖFFLER, 1984). They also differ distinctly from the intrusive-like melt pockets and pseudotachylite veins found in other shergottites such as ALHA 77005 and EETA 79001 (MCSWEEN *et al.*, 1979; MCSWEEN and STÖFFLER, 1980; MCSWEEN and JAROSEWICH, 1983). However, they may be genetically related to these intrusive-like veins and pockets of melts in the sense that they represent the incipient stage of this type of melt formation (see below). An indication for this interpretation is provided by the occurrence of melt regions in thin section ,34 where such regions are arranged in a linear array.

The type of localized *in situ* melting observed in Shergotty is known from shocked lunar basalts, *e.g.* from basalt 12057, in which plagioclase is partially converted to diaplectic plagioclase glass (ENGELHARDT *et al.*, 1971). In thin section 12057,14 greenish to brown melts are present in localized areas of about 1 mm^2 where large ilmenite grains are in contact with pyroxene and maskelynite (see Fig. 3 of ENGELHARDT *et al.*, 1971). These melts are also very heterogeneous in composition and represent a mixture of pyroxene, plagioclase, and ilmenite in variable proportions, very much like the Shergotty melts. Similar melts have been observed in lunar basalt 79155 which also contains partially isotropic plagioclase (SCHAAL and HÖRZ, 1977).

QUANTITATIVE DETERMINATION OF THE EQUILIBRIUM SHOCK PRESSURE

The only minerals which can be used for a quantitative shock pressure calibration of Shergotty are plagioclase and quartz. The most sensitive parameter for this calibration is the refractive index. From shock recovery experiments with single crystals and polycrystalline plagioclase a well-founded and rather exact relation between shock pressure and refractive index has been established for a wide range of plagioclase compositions (MILTON and DECARLI, 1963; BELL and CHAO, 1970; STÖFFLER and HORNEMANN, 1972; STÖFFLER, 1974; GIBBONS and AHRENS, 1977; OSTERTAG, 1983). In addition, the observed shock pressures required for partial and total isotropization of plagioclase can be applied for calibration. Such data are available from the shock recovery experiments mentioned above and from corresponding experiments on basalts (KIEFFER *et al.*, 1976; SCHAAL and HÖRZ, 1977). Finally, the relation between refractive index and degree of isotropization has been evaluated from

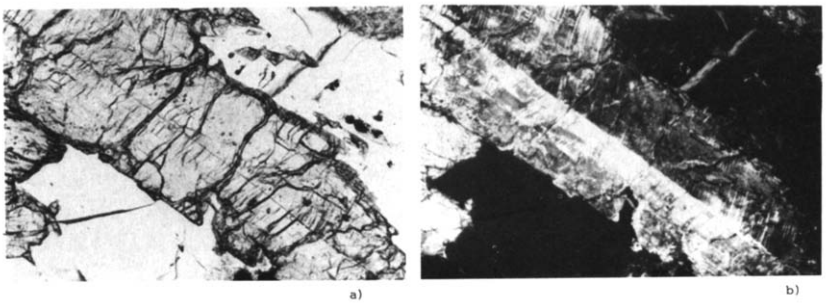


FIG. 3. Prismatic clinopyroxene twinned on (100) with shock-induced irregular and planar fractures and polysynthetic mechanical twin lamellae subparallel to (001); note preservation of fine structures in maskelynite (white) which displays rare concoidal fractures; a) transmitted light b) crossed polarizers; width of field: 1.17 mm; thin section ,34.



FIG. 4. Shock-induced deformation bands and polysynthetic,mechanical twin lamellae in clinopyroxene of Shergotty thin section ,30; crossed polarizers; width of field: 0.47 mm.

naturally shocked, plagioclase- and quartz-bearing rocks (STÖFFLER, 1967, 1971b; DWORAK, 1969). All available calibration data have been reviewed recently by STÖFFLER (1984).

The calibration data for plagioclase are plotted in Figs. 11 and 12. From the experimental data of Fig. 11 a best estimate of the equilibrium shock pressure for Shergotty (in the case of a single shock) can be derived assuming that the Shergotty labradorite is just beyond the transition from partially isotropic to completely isotropic (see section on SHOCK METAMORPHISM). For the given compositional range of the Shergotty maskelynite the pressure range is 29–31 GPa for complete isotropization. Observed relics of birefringent plagioclase could be best explained by an actual equilibrium shock pressure of 30 GPa leaving the possibility for some $<An_{50}$ grains to remain weakly birefringent.

This pressure estimate can be tested independently by using the calibration curves which relate the refractive index to the shock pressure (Fig. 12). Data for the refractive index of Shergotty maskelynite are available from many authors and are listed in Table 7. The data we obtained on the new Shergotty subsample ,1 fall within the range of data measured on other parts

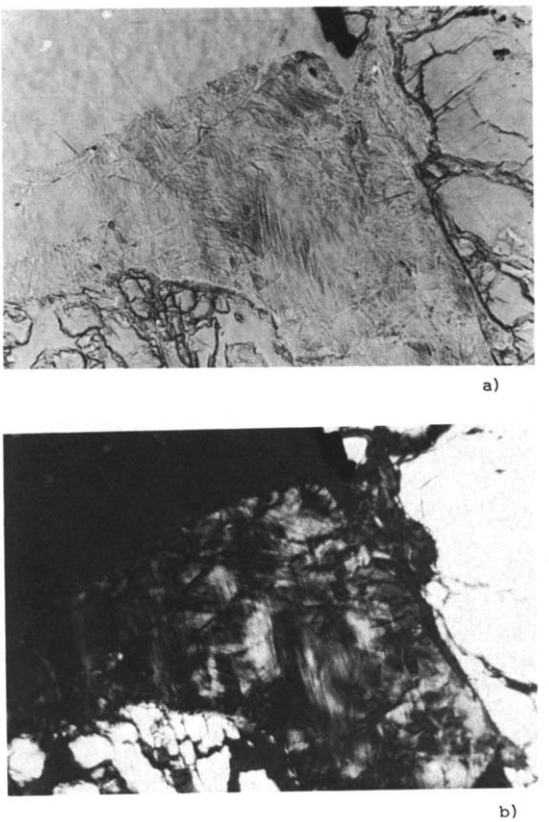


FIG. 6. Quartz with planar deformation structures and weak birefringence interstitial between maskelynite (upper part) and clinopyroxene (right and lower part) in Shergotty ultrathin section: a) transmitted light; b) crossed polarizers; width of field: 0.47 mm.

of the meteorite by previous authors (Table 7). The total range of the refractive indices and compositions of Shergotty maskelynite is plotted in Fig. 12. The reported maximum range of An-content is An_{36}–An_{60} (SMITH and HERVIG, 1979; STOLPER and MCSWEEN, 1979; SIMON *et al.*, 1985; DUKE, 1968; and this paper). The best average is An_{49} which is based on wet chemical

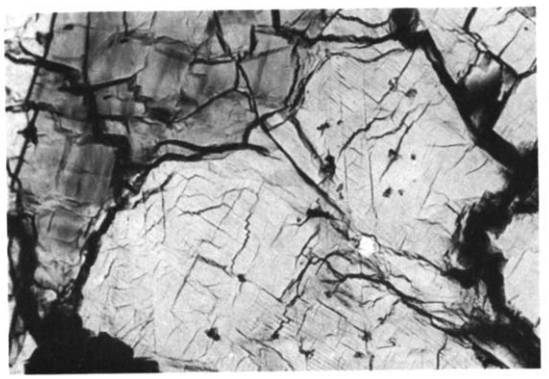


FIG. 5. Mechanical twinning (NW-SE), planar fractures (N-S and NE-SW) and irregular fractures in clinopyroxene of Shergotty thin section ,30; transmitted light; width of field: 1.17 mm.

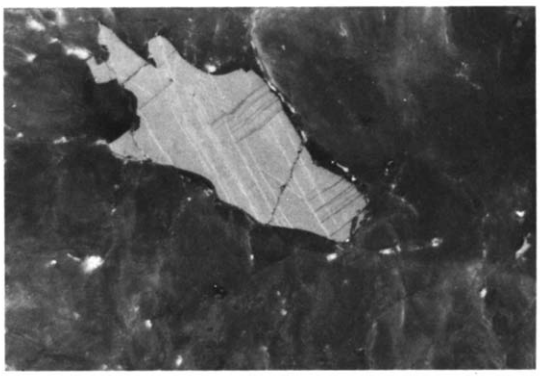
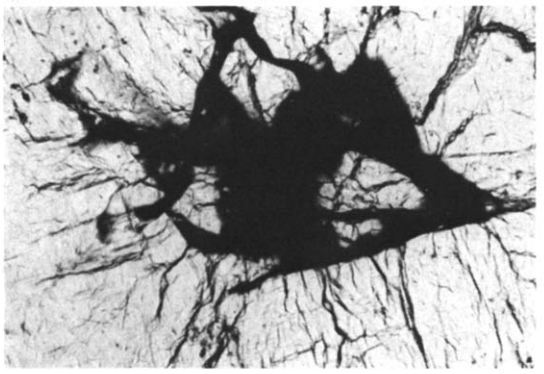
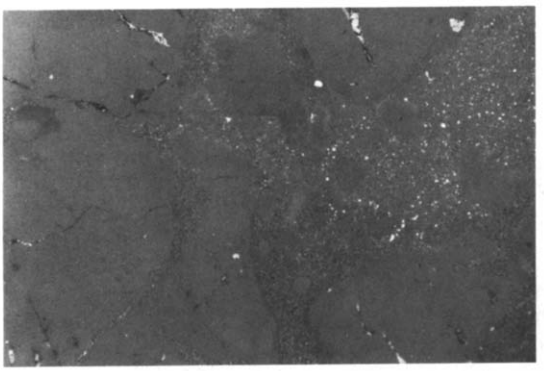


FIG. 7. Shock-induced mechanical twinning in ilmenite of Shergotty thin section ,34; reflected light; width of field: 0.29 mm.



a)



b)

FIG. 8a. Melt formed within clinopyroxene of Shergotty thin section ,30; transmitted light; width of field: 0.73 mm.

FIG. 8b. Enlarged view of Fig. 8a in reflected light showing fine-grained crystalline melt with exsolved pyrrhotite globules (white) and partially molten clinopyroxene (light gray); width of field: 0.29 mm.

analysis (DUKE, 1968). The published refractive indices range from 1.539–1.555. Reported averages are 1.546 (DUKE, 1968), 1.547 (LAMBERT and GRIEVE, 1984), and 1.5467 ± 0.0003 (this paper). Unfortunately, an exact correlation between refractive index and An-content has not been established for individual grains.



FIG. 9. Incompletely molten titanomagnetite within fine-grained crystalline melt of Shergotty thin section ,34; reflected light; width of field: 0.29 mm.

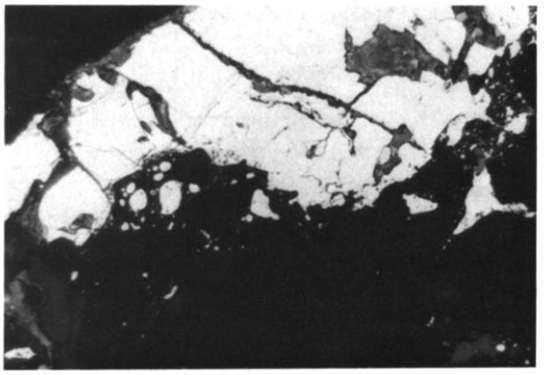


FIG. 10. Pyrrhotite grain (white) partially molten at lower rim with exsolved pyrrhotite globules (white) within fine-grained crystalline melt (dark) of Shergotty thin section ,30; reflected light; width of field: 0.29 mm.

However, it appears most plausible that high refractive indices correlate with maskelynite grains of high An-content. The assumption of a direct proportionality of refractive index and An-content is supported by the fact that the average An-content (An_{49}) and the average refractive index (1.5467) fall almost exactly on a straight line connecting the two data points which were obtained by plotting the lowest (highest) measured refractive index *versus* the lowest (highest) measured An-content (Fig. 12). The average falls on the 28 GPa isobar of Fig. 12, a value which is somewhat lower than the best estimate of 30 GPa from the data of Fig. 11. In Fig. 12, however, the average and the complete range of measured refractive indices for Shergotty maskelynite fall into a field between completely isotropic plagioclase (diaplectic glass) and still birefringent crystalline plagioclase (diaplectic crystals). This field has been defined on the basis of all available literature data. In particular, the data on An_{24} (STÖFFLER, 1974), An_{31} (STÖFFLER, 1967), An_{51} (STÖFFLER, 1974) and An_{55} (DWORAK, 1969) where refractive index measurements of birefringent and isotropic plagioclase of the same rock or single crystal are available, are crucial to the definition of this field. The data of these authors show that for a particular plagioclase composition the change of refractive index from the isotropic state to the anisotropic state is discontinuous. This means that in all observed cases (experimentally shocked An_{24} and An_{51} and naturally shocked An_{31} and An_{55}) the refractive index jumps from a low value where the plagioclase is isotropic to a high value at which it displays the lowest observable birefringence. In other words, a refractive index gap (which has not been discussed by previous workers) exists between still birefringent and isotropic shocked plagioclase as indicated in Fig. 12. The isotropic Shergotty labradorite falls into this gap.

In previous interpretations of a somewhat more restricted data base for shocked plagioclases a distinct boundary between birefringent diaplectic crystals and isotropic diaplectic glasses in the refractive index *versus* An-content relation was assumed (Fig. 4 of STÖFFLER, 1974; Fig. 4 of LAMBERT and GRIEVE, 1984). On the

Table 6: Chemical composition of in-situ melt products in Shergotty thin section ,30; n = number of analyses; σ = standard deviation

	1				2				3			
Oxide	min.	average	max.	σ	min.	average	max.	σ	min.	average	max.	n
SiO_2	37.3	44.5	50.0	3.9	29.2	32.3	41.4	5.3	46.1	47.6	49.4	3.0
Al_2O_3	4.0	7.30	11.3	2.6	1.05	2.26	4.13	1.19	7.1	8.80	9.7	0.8
FeO	13.8	20.90	28.2	4.5	28.6	34.70	43.5	4.9	18.6	19.60	21.1	0.9
MgO	3.0	5.20	7.6	1.6	1.22	1.91	2.87	0.50	7.3	8.00	10.1	0.9
MnO	0.32	0.49	0.67	0.12	0.49	0.60	0.72	0.08	0.4	0.45	0.5	0.05
CaO	9.2	12.48	18.9	2.5	3.8	6.70	9.1	2.3	7.4	7.80	8.0	0.2
Na_2O	0.73	1.72	3.02	0.72	0.59	0.89	1.49	0.34	1.56	1.99	2.32	0.22
K_2O	0.05	0.22	0.35	0.08	0.15	0.42	0.68	0.18	0.13	0.19	0.25	0.04
TiO_2	0.19	2.17	8.49	2.6	6.7	14.30	21.4	5.8	0.09	0.16	0.24	0.06
Cr_2O_3	-	0	-		0.09	0.15	0.23	0.05	0.13	0.18	0.26	0.04
P_2O_5	0.89	3.78	9.01	2.0	2.69	4.08	5.67	1.18	0.11	0.25	0.29	0.05
SO_3	0.19	0.72	1.26	0.48	2.01	2.63	3.92	0.61	2.21	3.27	4.14	0.56
Cl	0.0	0.16	0.25	0.07	0.06	0.12	0.17	0.05	0.0	0.01	0.05	0.02
Total		99.64				101.06				98.30		
n		14				8				9		

basis of this assumption, it was thought that the whole range of refractive indices of Shergotty maskelynite would be within the field of diaplectic crystals and therefore would be higher than the refractive indices of all known diaplectic labradorite glasses recovered from experimentally and naturally shocked rocks (STÖFFLER, 1974; GIBBONS and AHRENS, 1977; LAMBERT and GRIEVE, 1984). As a consequence of this interpretation the hypothesis was suggested that the Shergotty maskelynite must have been densified by a second shock event (GIBBONS and AHRENS, 1977; LAMBERT and GRIEVE, 1984). Such a shock densification process was demonstrated experimentally on various glasses (ARNDT *et al.*, 1971; GIBBONS and AHRENS, 1971) including terrestrial maskelynite (An_{50}) of Mistastin Lake impact crater (LAMBERT and GRIEVE, 1984).

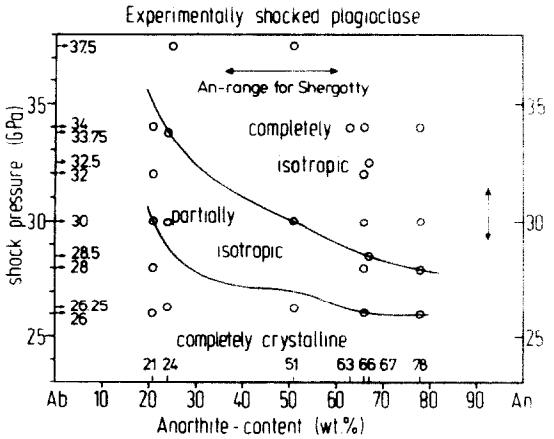


FIG. 11. Shock pressure required for partial and total isotropization of plagioclase as a function of An-content of the plagioclase; measured An-range for Shergotty is given for comparison; data from shock experiments of STÖFFLER (1974), OSTERTAG (1983), BELL and CHAO (1970), and GIBBONS and AHRENS (1977) as indicated by the arrows; circles represent individual experiments.

We have carefully checked the data basis for this densification hypothesis. The following important conclusions must be drawn:

(1) Single crystal labradorite of Shergotty composition converts to completely isotropic maskelynite at 30 GPa as indicated also by the IR-spectra of An_{51} (Fig. 14 of STÖFFLER, 1974). Therefore, the statement of STÖFFLER and HORNEMANN (1972) that An_{51} is still a disordered diaplectic crystal at 30 GPa is incorrect.

(2) The boundary between birefringent diaplectic An_{51}-crystals and isotropic diaplectic labradorite (An_{51}) glass is not defined by the refractive index of 1.544 or 1.538 as assumed in Figs. 4 of STÖFFLER (1974) and LAMBERT and GRIEVE (1984), respectively; instead the lowest refractive index of any known birefringent An_{51}-crystal is 1.5516 and the highest refractive index of any known An_{51}-diaplectic glass is 1.5419 (STÖFFLER, 1974; Figs. 12 and 13).

(3) Consequently, the average refractive index of 1.5467 for Shergotty maskelynite (An_{49}, average) does not overlap with the refractive index of diaplectic crystals of approximately the same composition found in experimentally (An_{51}) or naturally shocked labradorite (Manicouagan, An_{55}) but rather is slightly higher than the highest refractive index of experimentally produced diaplectic An_{51}-glass which is 1.5419 (STÖFFLER, 1974; see also DWORAK, 1969). These relations are shown in Fig. 13.

On the basis of the facts outlined above, the arguments of LAMBERT and GRIEVE (1984) to postulate that the Shergotty meteorite experienced a "second shock event in the range 10–20 GPa, following an early shock event at 30–50 GPa" are marginal and partially invalid. On the other hand, this hypothesis cannot be completely ruled out at this point, but the problems with the hypothesis of Lambert and Grieve are the following: Firstly, the existence of weakly birefringent quartz and of remnants of birefringent plagioclase clearly indicate that the first shock was on the order

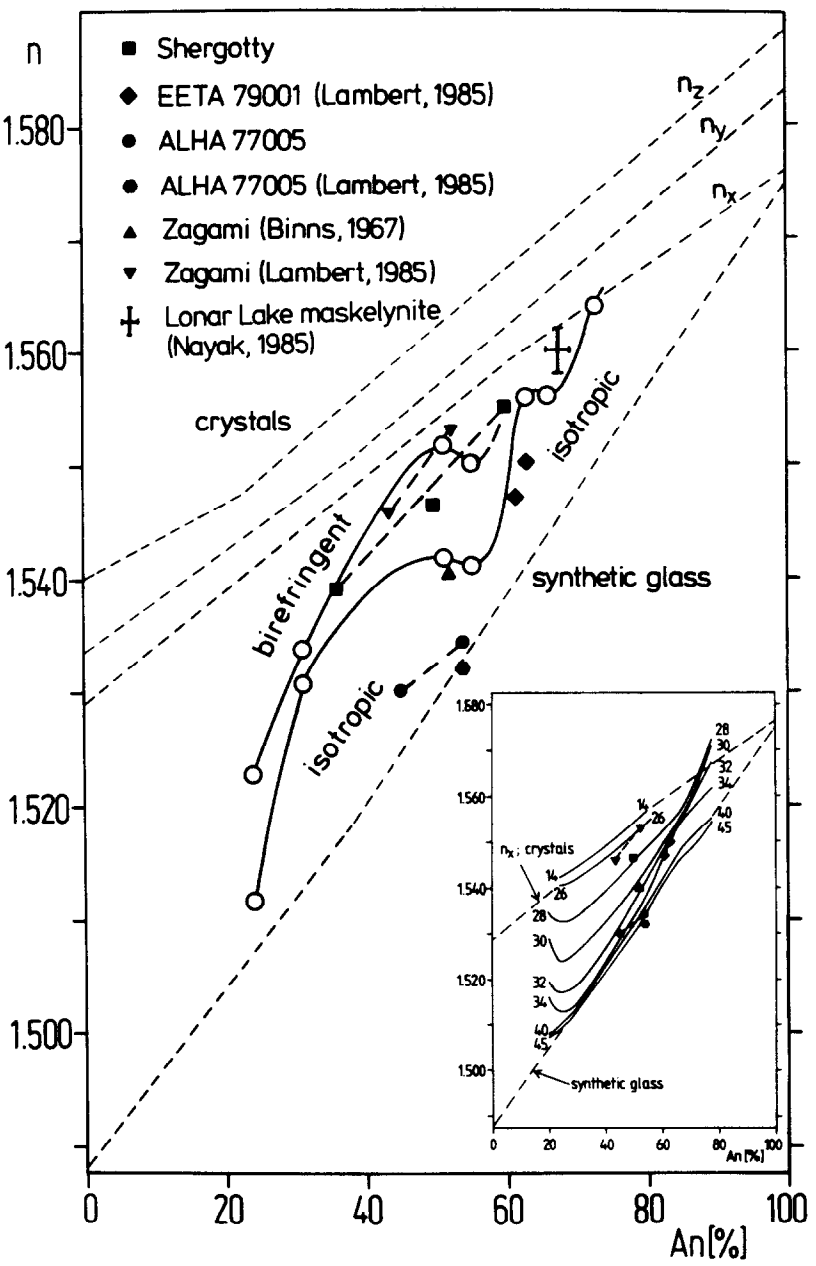


FIG. 12. Refractive indices *versus* An-content for unshocked (n_x, n_y, n_z), shocked, and fused plagioclase (synthetic glass); thick solid lines separate fields for shocked, birefringent plagioclase (diaplectic crystals) and shocked, isotropic plagioclase (diaplectic glass = maskelynite); these fields are based on data from shock experiments and from naturally shocked plagioclase as indicated by circles which represent the lowest measured refractive index of diaplectic crystals and the highest measured refractive index of diaplectic glass (data from STÖFFLER, 1967, 1974, for An_{24}, An_{31}, and An_{51}; from DWORAK, 1969, for An_{55}; from GIBBONS and AHRENS, 1977, for An_{63}; from OSTERTAG, 1983, for An_{66} and An_{78}); thick dashed lines connect minimum and maximum values for Shergotty and ALHA 77005 maskelynite with average given for Shergotty (see text); inset at lower right corner for shock pressure calibration; isobars are based on experiments by STÖFFLER and HORNEMANN (1972), STÖFFLER (1974), and OSTERTAG (1983).

of 30 GPa; these phases simply would not exist at pressures above 30 GPa (Figs. 11 and 14), and they certainly cannot be produced by a second shock of less than 20 GPa. Secondly, the equilibrium peak pressure of a second shock which must be capable of densifying regular 30 GPa-diaplectic glass of An_{49} composition in such a way that the refractive index increases from about 1.542 to the observed value of 1.547, can only be either in the 14 ± 2 GPa or in the 22 ± 1 GPa range (Fig. 13) allowing for the uncertainties of the densification curve of LAMBERT and GRIEVE (1984). All other shock pressures would either produce a higher or lower densifi-

Table 7: Refractive index and composition of Shergotty maskelynite; n = number of measured grains

Refractive index range	Refractive index average (n)	Composition range	Composition average	(Authors) References
	1.5482		An_{48}	Michel, 1912; Tschermak, 1872
	1.547		An_{50}	Bunch et al., 1967
	1.544		An_{51}	Binns, 1967
1.543 - 1.555 ± 0.002	1.546 (14)	An_{46} - An_{54}	An_{49}	Duke, 1968
1.539 - 1.553 ± 0.0005	1.547 (7)			Lambert and Grieve, 1984
1.5456 - 1.5490 ± 0.0003	1.5467 (20)	An_{44} - An_{54}		this paper
		An_{41} - An_{56}		Smith and Hervig, 1979
		An_{43} - An_{52}		Stolper and McSween, 1979
		An_{36} - An_{54}		Simon et al., 1985 and this paper
1.539 - 1.555	1.547	An_{36} - An_{60}	$An_{49}(Or_{2.5}Ab_{48}An_{49})$	

last line gives observed total ranges and averages of all available data

cation than the one required for producing an An_{49}-glass with a refractive index of 1.547, the average value of the Shergotty maskelynite (Fig. 13). These limitations for the peak pressure of the second shock event (either 14 or 22 GPa but not just any pressure between 10 and 20 GPa as postulated by LAMBERT and GRIEVE, 1984) would impose very special conditions for the Shergotty shock history which can be considered unlikely.

Therefore, the question remains whether the observed shock effects in Shergotty could be still explained by a single shock event. All quantitative data which are directly relevant to this question, namely the data on experimentally shocked labradorite (An_{51}) and on naturally shocked Mistastin (An_{50}) and Manicouagan (An_{55}) labradorites (STÖFFLER, 1974; DWORAK, 1969; LAMBERT and GRIEVE, 1984) have been obtained on either single crystals (An_{51}) or nearly monomineralic anorthosite (An_{50} and An_{55}). However, in Shergotty the labradorite coexists with clinopyroxene as the main constituent. The best experimentally and naturally shocked analogs for Shergotty are the basalt from Lonar Lake impact crater, India (KIEFFER *et al.*, 1976) and two experimentally shocked basalts, the Lonar Lake basalt and the lunar basalt 75035 (KIEFFER *et al.*, 1976; SCHAAL and HÖRZ, 1977) which are rather comparable in mineralogy and texture to Shergotty. Shock experiments were made in the 20–80 GPa range (KIEFFER *et al.*, 1976; SCHAAL and HÖRZ, 1977). At 27.0 GPa 90% of the labradorite is isotropic according to Table 3 of KIEFFER *et al.* (1976), at 33.4 GPa already traces of plagioclase "melt" are recorded. Compared with the corresponding effects in shocked single crystal labradorites (An_{51}, An_{63}, and An_{66} of STÖFFLER (1974), GIBBONS and AHRENS (1977), and OSTERTAG (1983), respectively), it appears that the threshold pressure for certain shock transformations such as isotropization and melting are lower in the polycrystalline, pyroxene-rich rocks than in the single crystals, *e.g.* at 28.7 GPa no glass is recorded in a An_{63} single crystal by GIBBONS

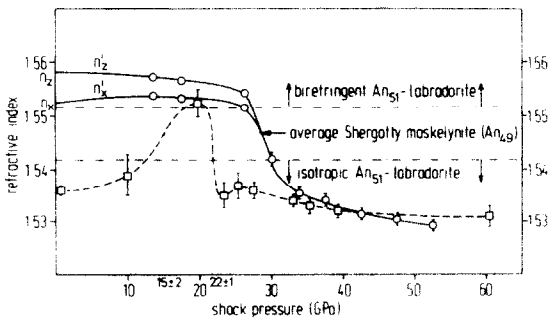


FIG. 13. Refractive indices *versus* shock pressure of shocked labradorite (An_{51}; open circles; data from STÖFFLER, 1974) and reshocked maskelynite (An_{50}; open squares; data from LAMBERT and GRIEVE, 1984); average refractive index of Shergotty maskelynite (arrow) would require either 15 or 22 GPa shock pressure if produced by shock densification of preexisting maskelynite.

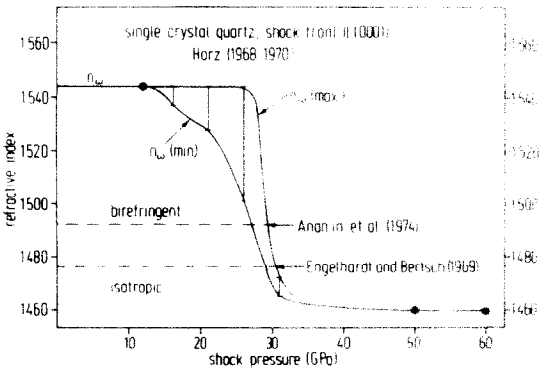


FIG. 14. Refractive index (n_ω) *versus* shock pressure of shocked quartz; data for 50 and 60 GPa from MÜLLER and HORNEMANN (1967) and DECARLI and JAMIESON (1959), respectively; dashed-dotted lines indicate lowest refractive index of birefringent crystals according to two different authors.

and AHRENS (1977) whereas 90% of the Lonar basalt labradorite (An_{55}–An_{69}) is isotropic at 27.0 GPa (KIEFFER *et al.*, 1976). If we apply this observation to the shock pressure calibration of the Shergotty meteorite an *equilibrium shock pressure of 29 GPa* becomes most probable instead of 30 GPa derived from the data on single crystal experiments (Fig. 11).

A detailed determination of the refractive index of the shocked Lonar basalt labradorite would be most important. NAYAK (1985) reports an average refractive index of 1.560 ± 0.002 for a maskelynite phenocryst (An_{66-69}) from Lonar basalt. This value is indeed higher than the highest measured refractive index (n = 1.5563) of an experimentally shocked single crystal labradorite (An_{66}; OSTERTAG, 1983) and, similarly to the Shergotty maskelynite, falls outside of the field for isotropic plagioclase derived from data on single crystals and nearly monomineralic, feldspathic rocks (Fig. 12). This observation strongly supports the interpretation that a single shock can be responsible also for the high refractive index of the Shergotty maskelynite.

From the observed shock effects in quartz an equilibrium shock pressure of 29 ± 2 GPa can be derived (see data of Fig. 14) in full accordance with the pressure estimate based on maskelynite.

In summary, we recognize two possible scenarios for the shock history of Shergotty on the basis of the data discussed above:

(1) Shergotty was shocked in a single shock event of at least 29 ± 1 GPa equilibrium shock pressure with localized *in situ* melts formed simultaneously by very high stress and temperature concentration of some 60–80 GPa due to shock wave reverberations at grain contacts of low and high density minerals; or

(2) Shergotty was shocked by a first shock wave of at least 29 ± 1 GPa equilibrium peak pressure and subsequently reshocked either by a shock wave of about 14 ± 2 GPa or 22 ± 1 GPa equilibrium peak pressure by which the maskelynite formed at or above 29 GPa was slightly densified. The localized *in situ* melting would most probably be the result of the first shock.

We should emphasize that we favour scenario (1) for reasons discussed above, but we cannot completely exclude scenario (2) at this point as we have done earlier (OSTERTAG *et al.*, 1985), although scenario (2) appears to be highly improbable.

SHOCK RELATED THERMAL HISTORY

According to the principles of shock wave propagation in solids, any particular, transient shock stress is related to a specific transient shock temperature which drops to a lower post-shock temperature upon pressure release due to the non-isentropic nature of the shock process. The magnitudes of both shock and post-shock temperatures depend only on the Hugoniot data of the compressed solid at any given equilibrium shock pressure and initial temperature. Shock and post-shock temperature data for silicate minerals and rocks are available from calculations and experiments (*e.g.*, WACKERLE, 1962; AHRENS *et al.*, 1969; MASHIMO *et al.*, 1980; RAIKES and AHRENS, 1979; BOSLOUGH and AHRENS, 1983; see also review in STÖFFLER, 1984). The magnitude of the shock-induced heating is largely proportional to the compressibility of the solid (GRADY, 1980). More heat is induced in open crystal structures such as framework silicates than in dense structures such as mafic silicates. Consequently, an equilibrium temperature between the temperatures of labradorite and pyroxene will be developed immediately after shock decompression in the Shergotty meteorite. Therefore, the temperatures estimated for Shergotty (Table 8) which are mainly based on data for framework silicates (feldspar, quartz) will be upper temperature limits. They hold for an ambient pre-shock temperature of about 25°C.

The data of Table 8 indicate that the equilibrium post-shock temperature of Shergotty did not exceed 200°C, if the parent rock was shocked near the planetary surface at low initial temperature. For other shergottites temperatures of up to 800°C can be inferred from their degrees of shock. In regions of localized *in situ* melting large deviations from the equilibrium temperature occur, exceeding at least 1600°C which as a post-shock temperature would require a minimum peak shock of 60 GPa or more (see Fig. 9 of STÖFFLER, 1984). In fact, SCHAAL and HÖRZ (1977) observed melting of pyroxene in experimentally shocked basalts only above 80 GPa. It should be noted that in case of a second shock compression of a rock which already contains maskelynite the shock-induced temperatures (shock and post-shock) would be considerably higher than in the first shock because of the low density of maskelynite. Data for quartz and silica glass indicate a difference of about 800 and about 1000°C between the corresponding post-shock temperatures of both phases at 20 and 30 GPa, respectively. Correspondingly high shock temperatures were measured on anorthite glass (BOSLOUGH and AHRENS, 1983).

The theoretical shock-induced temperatures discussed above can be checked by data obtained through annealing experiments on maskelynite (DUKE, 1968; BELL and CHAO, 1970; ENGELHARDT *et al.*, 1970). In all naturally and experimentally shocked feldspars studied (An_{22}, An_{41}, An_{49}, An_{67}) the refractive index of diaplectic glass (maskelynite) decreased upon annealing at temperatures between 400 and 800°C and reached the value of the synthetic "melt" glass. The only exception from this behaviour was found with lunar diaplectic anorthite glass (An_{92}) which increased its refractive index in the 500–800°C range (ENGELHARDT *et al.*, 1970). However, the index of the unheated An_{92}-glass is already in the range of the corresponding crystalline anorthite in contrast to plagioclase with An < 75 (see Fig. 12). Above about 800°C diaplectic glasses increase their refractive index and convert rapidly to the crystalline phase between 900 and 1000°C (BUNCH *et al.*, 1968; ARNDT *et al.*, 1982).

Table 8: Calculated shock and post-shock temperatures in framework silicates and estimates for the shergottite meteorites; pre-shock temperature is 25° C; * = measured

Material	Shock pressure (GPa)	Shock temperature (°C)	Post-shock temperature (°C)	Authors
Quartz	20	-	255*	Raikes and Ahrens, 1979
	20	238	168	Wackerle, 1972
	30	336	214	
	50	1630	1160	
	60	2650	1920	
Silica	20	-	800	Gibbons and Ahrens, 1971
	30	-	1260	
	30	495	470	Wackerle, 1962
Oligoclase	33.6	892	-	Ahrens et al., 1969
	50.4	1774	-	
	63.4	2474	-	
	18.0	-	27 - 35	
	27.2	-	129 - 206	
	41.7	-	327 - 395	
Bronzite	14.8	-	160*	Raikes and Ahrens, 1979
	26.0	-	240*	
Olivine	34.7	177	-	Ahrens et al., 1976
	44.1	427	-	
	51.3	627	-	
	60.6	927	-	
	28.0	-	167*	Raikes and Ahrens, 1979
Shergotty	29 ± 1	~300	200 ± 20	this paper
Zagami	a)31 ± 2	~350	220 ± 50	this paper
	b)27	-	-	Lambert, 1985
EETA 79001	34 ± 1	~500	250 ± 50	this paper; based on pressure estimate of Lambert, 1985
ALHA 77005	a)43 ± 2	~800 - 1000(?)	400 - 800(?)	this paper
	b)45	-	-	Lambert, 1985

We conclude from the data discussed above that in the An_{20}–An_{75} range maskelynite decreases its refractive index and density more readily with increasing refractive index of the unannealed material. Shergotty maskelynite, which has the highest refractive index of any known diaplectic glass of the same composition, starts to anneal at 400°C (DUKE, 1968); the An_{22}, An_{41}, and An_{67} glasses of BELL and CHAO (1970) start to anneal at 525–650°C. The major consequence of this is, as already pointed out by others (DUKE, 1968; LAMBERT, 1983), that Shergotty could not have been heated after the shock metamorphic event to more than 400°C.

CONCLUSIONS AND IMPLICATIONS FOR THE SHOCK HISTORY OF SHERGOTTITES

From the optical properties of maskelynite an equilibrium shock pressure of 30 ± 1 GPa can be derived for the Shergotty meteorite if the pressure calibration is based on experimentally shocked single crystal plagioclase. Pressure calibration using data from experimentally shocked basalts indicate a somewhat lower, more preferable pressure of 29 ± 1 GPa. These pressure estimates are in accordance with shock effects observed in other mineral phases of Shergotty (quartz, clinopyroxene and ore minerals; HÖRZ *et al.*, 1986). The equilibrium post-shock temperature of Shergotty did not exceed 200 ± 20°C and any later reheating above 400°C can be excluded (DUKE, 1968). Locally, stress and temperature concentrations on the scale of individual mineral grains produced shock pressures in excess of 60 to 80 GPa and post-shock temperatures exceeding 1600–2000°C.

The refractive index of the Shergotty maskelynite falls into a gap between the refractive indices of experimentally and naturally shocked isotropic plagioclase (single crystals or polycrystalline andesine-labradorite) and the refractive indices of the corresponding shocked plagioclases with strongly reduced birefringence. This relatively high index of the isotropic Shergotty andesine-labradorite has been explained by shock-induced densification of preexisting maskelynite during a second shock event (LAMBERT and GRIEVE, 1984). We argue that a second shock is not required to produce the observed optical characteristics of the Shergotty maskelynite. Moreover, it would impose very narrow constraints on the peak pressure range allowed for the invoked densification and is therefore highly improbable.

A comparison of the shock histories of all known shergottites shows that the equilibrium shock pressure experienced by Shergotty, Zagami, EETA 79001, and ALHA 77005 increases from 29 ± 1 GPa to 31 ± 2 GPa, 34 ± 2 GPa, and 43 ± 2 GPa, respectively (Table 8; Fig. 12; see also LAMBERT, 1985). The estimates for Zagami and EETA 79001 are based on refractive index

data of maskelynite measured by BINNS (1967) and LAMBERT (1985), respectively. LAMBERT (1985) estimates 27 GPa for the Zagami shergottite from a very high refractive index which deviates appreciably from the value given by BINNS (1967). This low shock pressure for Zagami is somewhat difficult to reconcile with the fact that the Zagami plagioclase is completely isotropic. Remnants of birefringent plagioclase should be expected at this pressure based on the experimental data discussed in the previous sections. The type and amount of localized whole rock melting in each of these meteorites appears to be consistent with an increasing equilibrium shock pressure in the sequence Shergotty, Zagami, EETA 79001, ALHA 77005. The very localized *in situ* melting observed in Shergotty may be interpreted as an incipient stage in the formation of pseudotachylite-like melt veins commonly found in all other shergottites and intrusive-like larger melt veins and pockets observed in EETA 79001 and ALHA 77005 (MCSWEEN and JAROSEWICH, 1983; MCSWEEN and STÖFFLER, 1980). The latter may represent the most advanced stage of localized whole rock melting in shergottites. It should be emphasized that this is a very tentative interpretation. More detailed studies are needed to understand fully the shock histories of the shergottites. On the other hand, there is clear evidence from observations of terrestrial and lunar impact-metamorphosed rocks that the equilibrium (maskelynite formation) and non-equilibrium shock effects (localized melting) observed in the shergottites can be produced by a single shock event. Pseudotachylite-like melt veins in combination with partial to total isotropization of framework silicates are very common in shocked terrestrial and lunar rocks (STÖFFLER *et al.*, 1979). They occur in rock fragments of allochthonous polymict breccias and in shocked rocks of the crater basement. The melt pockets of the shergottites do not have direct analogues in terrestrial impact formations although mm to cm thick layers of whole rock melts formed *in situ* within very strongly shocked rocks of granitoid composition are well-known from the Ries and Haughton Dome craters (STÖFFLER *et al.*, 1979; OSTERTAG *et al.*, 1984).

The estimated equilibrium shock pressures of the various shergottites can be used to derive the particle velocities to which the shocked materials were accelerated. If the specific effects at the free target surface on the particle motion are neglected, basaltic rocks affected by shock pressures in the 29 to 43 GPa range will move with particle velocities of 1.5 to 2.0 km/s during shock compression according to measured Hugoniot data of basalts (see compilation in STÖFFLER, 1982). At a free surface the material will be accelerated to approximately twice the particle velocity (free surface approximation) leading to ballistic ejection velocities of about 3 to 4 km/s. These velocities are much less than the escape velocity of a planet like Mars. As discussed by several authors (*e.g.* NYQUIST, 1983, 1984; MELOSH, 1984) special ejection mechanisms are therefore required to accelerate the moderately shocked shergottites to velocities by which an escape from the Martian gravity field would become possible.

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